

Roll No. :

MID SEMESTER EXAMINATION MARCH 2025

Name of the Program: **B.Tech**

Semester: **8**

Name of the Course: **Digital System using VHDL**

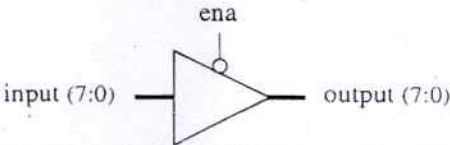
Course Code: **TEC 853**

Time : 90 Minutes

Maximum Marks : 50

Note:

- i. Answer all the questions by choosing any one of the sub questions
- ii. Each questions carries 10 marks

Q1	(10 Marks)	
(a)	Discuss the VLSI Design Flow with the help of flowchart.	CO1
	OR	
(b)	Explain the basic structure of VHDL code.	CO1
Q2	(10 Marks)	
(a)	Discuss the various data types of VHDL with suitable examples.	CO1
	OR	
(b)	Explain the pre-defined operators in VHDL with examples.	CO1
Q3	(10 Marks)	
(a)	Explain Signal Attributes and Data Attributes in VHDL.	CO1
	OR	
(b)	Write VHDL code for 4×1 Multiplexer using :- 1. Operators 2. WHEN statement	CO2
Q4	(10 Marks)	
(a)	What is the use of the following statements:- LIBRARY IEEE; USE IEEE.STD_LOGIC_1164.ALL;	CO1
	OR	
(b)	Write VHDL code for the following Tri-state buffer 	CO2
Q5	(10 Marks)	
(a)	Write the VHDL code for 8-bit input encoder.	CO2
	OR	
(b)	Write VHDL code of JK Flip-Flops.	CO2

